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New paleomagnetic pole and magnetostratigraphy of Faroe Islands flood volcanics, North Atlantic igneous province

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Abstract

A paleomagnetic sampling was carried out along four sections (altogether 86 lava flows, 548 samples) in the North Atlantic Igneous Province outcropping in Faroe Islands, Denmark. The four polarity zones in the 700-m-thick exposed part of the Faroes lower formation can be correlated with the geomagnetic polarity time scale as C26n–C25r–C25n–C24r. The seven lava flows erupted during C25n indicate a very low eruption rate in the upper part of the Faroes lower formation of ~1/70 kyr. The Faroes middle and upper formations (composite thickness ~2300 m) are all reversely magnetized corresponding to C24r. The eruption rate at the onset of middle formation volcanism was very high as evidenced by several thick lava sequences recording essentially spot readings of the paleomagnetic field. The shift in eruption rate between the Faroes lower and middle formations and evidence that onset of the Faroes middle formation volcanism took place in C24r are of particular importance, placing onset of middle formation volcanism in close temporal relation to North Atlantic continental break-up and the late Paleocene thermal maximum. After grouping flows recording the same field directions, we obtained 43 independent readings of the paleomagnetic field, yielding a paleomagnetic pole with coordinates 71.4°N, 154.7°E (A_{95} = 6.0°, K = 14, N = 43); age 55–58 Ma. The pole is supported by a positive reversal test. Paleosecular variation, estimated as the angular standard deviation of the virtual geomagnetic pole distribution 21.7°+3.9°/−2.8°, is close to expected for the given age and paleolatitude. Our new Faroes paleomagnetic pole is statistically different from the majority of previously published poles from the British and Faroes igneous provinces, and we suggest that these older data should be used with care. © 2002 Elsevier Science B.V. All rights reserved.

Keywords: North Atlantic; Faeroe Islands; Paleogene; paleomagnetism; magnetostratigraphy

1. Introduction

With an estimated total volume of ~10^7 km^3 [1] the North Atlantic Igneous Province (NAIP, Fig. 1a) is one of the largest magmatic events on Earth in the last 200 Myr. Accurate dating of the NAIP has been the focus of many geochronology papers in the last 20–30 yr in order to constrain the temporal evolution of the flood volcanism and

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elucidate its relationship to the Iceland mantle plume, the North Atlantic continental break-up, and late Paleocene–early Eocene global environmental changes. The Faroe Islands take up a central position within the NAIP (Fig. 1) representing one of the thickest and stratigraphically most complete volcanic sections. In this study, we present new magnetostratigraphy data from the Faroes. The magnetostratigraphic age constraints are particularly important for correlating NAIP volcanism with other geological events dated by the geomagnetic polarity time scale (GPTS), such as onset of North Atlantic sea-floor spreading [2,3] and the late Paleocene thermal maximum [4,5].

Another major aim of this study was to obtain a new reliable paleomagnetic pole for the Faroe Islands. Although the Eurasian part of the NAIP has already been extensively studied, it is worth noticing that all 27 previously published paleomagnetic poles are based on pre-1980 laboratory techniques, i.e. based on either natural remanent magnetization (NRM) values or single step blanket demagnetization or the outdated minimum dispersion method. Although some authors have deemed pre-1980 paleomagnetic data reliable [6], others have questioned their reliability [7]. For example, a review of the large paleomagnetic data set from the ~65 Myr old Deccan traps revealed a discrepancy between pre- and post-1980 data which was ascribed to insufficient removal of viscous overprints in the earlier studies [8]. A more direct critique arises from the significantly different results obtained from young volcanic rocks when pre-1980 studies were repeated using modern laboratory techniques [9–11]. It is therefore unfortunate that recent analysis of the geometry of the Earth's paleomagnetic field [12,13], and apparent polar wander [14] as well as true polar wander [15] to a certain extent relies on pre-1980 paleomagnetic directional data, including Eurasian NAIP data. We present here the first pole for the Eurasian NAIP that fulfills present-day standards of laboratory and analytical techniques. In the following we pay particular attention to differences between our new paleomagnetic pole and the large pre-1980 data set.

2. Geology and sampling

The Faroe Islands are situated in the northern part of the Faroe–Rockall Plateau, which forms an extensive area of elevated sea-floor to the north and west of the British Isles (Fig. 1a). The plateau is almost completely covered with subaerially erupted basalts, now submerged, and land exposures of basalt are confined to the Faroe Islands. The Faroe–Rockall Plateau is underlain by continental crust [16,17] with only minor extension since basalt emplacement [18] and hence in a stable plate tectonic position in relation to Eurasia. The structural attitude of strata on and around the Faroes, generally less than 4°, is dominated by two uplift domes, a large one centered NW of the islands and a smaller southern one. The main phase of doming seems to have occurred after the extrusion of the main lava sequence and is probably related to the formation of master joints, minor faults, dikes and sills [19]. The total thickness of the volcanic pile at the Faroe Islands is more than 6.5 km, the lower 3.5 km of which is only known from drill hole Lopra-1 [20] and its recent extension (unpublished). The regional basalt stratigraphy is well established and informally divided into a lower, middle and upper formation (Fig. 1b; [19,21,22]). The lower formation consists of fairly uniform, aphyric to sparsely plagioclase–phyric basaltic lava flows with an average thickness of about 20 m. The middle formation starts with pyroclastic

→

Fig. 1. Maps of (a) the NAIP and (b) Faroe Islands (after [3,21,22]). The locations of the profiles sampled in this study are shown as squares. The exact geographic coordinates are given in Tables 1a and 1b. The position of the four paleomagnetic profiles in the lithostratigraphy is indicated where 1 is for Vágur, 2 for Fámjin, 3 for Sandvík and 4 for Breidadalsá. (A) horizon comprises coal-bearing sediment sequence, irregular intrusives and tuffs. (B) is an auxiliary mapping horizon. (C) horizon is the boundary between the middle and upper formations, about which level several basaltic sills are intruded.

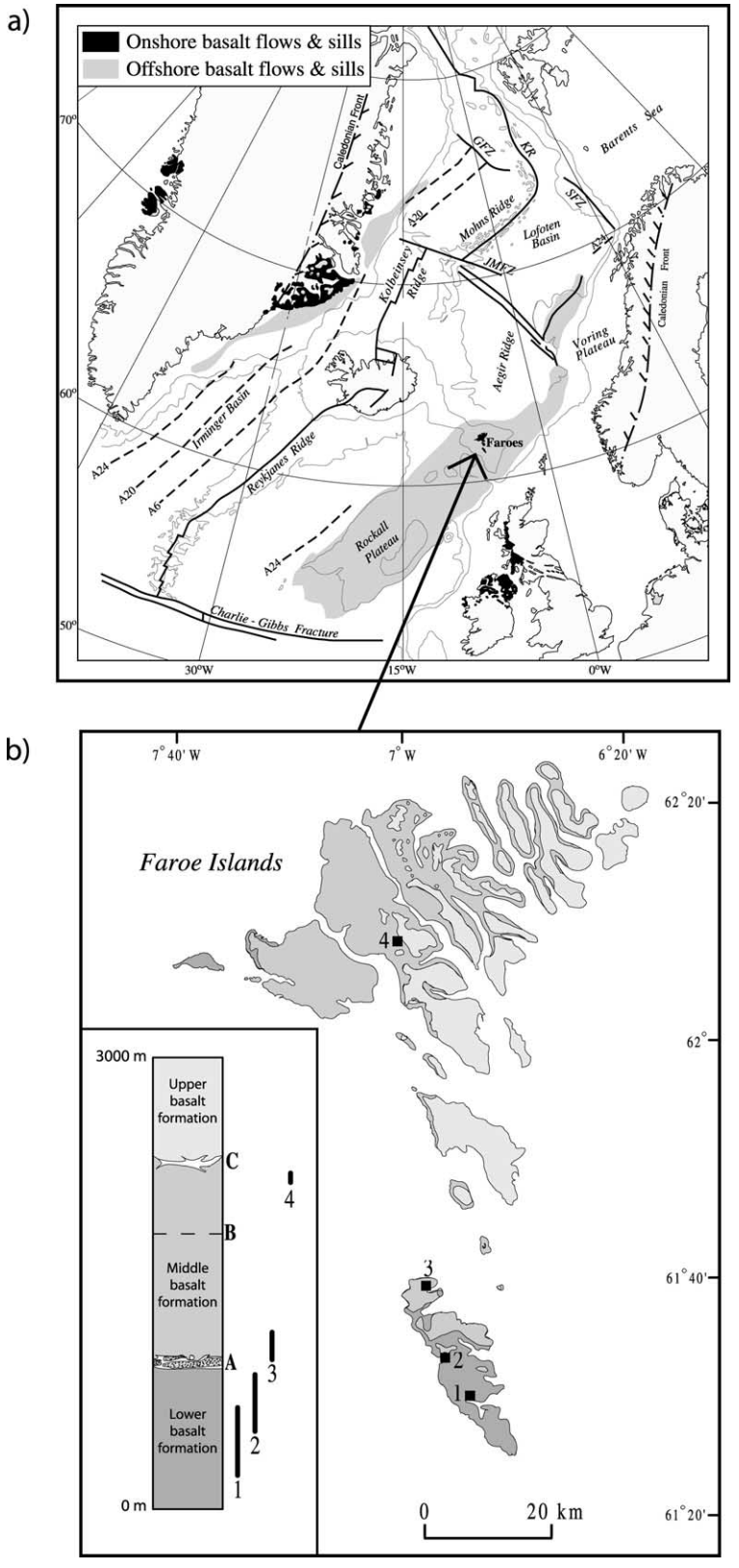

a)
Onshore basalt flows & sills
Offshore basalt flows & sills
Caledonian Fault
GFZ
KR
Barents Sea
SFZ
A20
Mohns Ridge
Lofoten Basin
JMFZ
Kolbeinsey Ridge
Caledonian Front
Aegir Ridge
Voring Plateau
Irminger Basin
A24
A20
A6
Reykjanes Ridge
Faroes
Rockall Plateau
Charlie - Gibbs Fracture
70°
60°
50°
30°W
15°W
0°W
b)
7° 40' W
7° W
6° 20' W
62° 20'
62°
61° 40'
61° 20'
Faroe Islands
4
3
2
1
0
20 km
3000 m
0 m
Upper basalt formation
Middle basalt formation
Lower basalt formation
C
B
A

Table 1a
Paleodirectional data for the DGs of the middle formation

DG #	Site	Alt. (m)	*n*/*N*	Inc. (°)	Dec. (°)	*k*	α_{95} (°)	VGP Lat. (°N)	VGP Long. (°E)
Sandvík									
1	S1	257	4/4	−52.8	142.7	554	3.9		
DG1 mean			**4/4**	**−52.8**	**142.7**	**554**	**3.9**	**−53.1**	**50.5**
2	S2	249	4/4	−48.1	145.3	999	2.9		
DG2 mean			**4/4**	**−48.1**	**145.3**	**999**	**2.9**	**−50.3**	**44.2**
3	S3	246	3/3	−65.7	239.5	959	4.0		
3	S4	241	2/2	−62.3	233.0				
3	S5	237	4/4	−66.5	229.6	2637	1.8		
3	S6	233	4/4	−65.4	234.8	998	2.9		
3	S7	231	5/5	−63.4	233.1	660	3.0		
3	S8	227	4/4	−65.4	233.8	1008	2.9		
3	S9	223	6/6	−63.1	233.8	254	4.2		
3	S10	219	4/4	−62.5	244.0	2294	1.9		
3	S11	216	5/5	−65.2	236.1	1094	2.3		
3	S12	213	2/2	−65.6	240.9				
3	S13	210	5/5	−64.1	238.1	800	2.7		
3	S14	208	4/4	−65.7	234.1	257	5.7		
DG3 mean			**48/48**	**−64.6**	**235.7**	**529**	**0.9**	**−55.3**	**262.8**
4	S15	205	5/5	−50.0	229.5	419	3.7		
4	S16	200	6/6	−50.9	222.5	628	2.7		
DG4 mean			**11/11**	**−50.5**	**225.8**	**383**	**2.3**	**−47.7**	**287.5**
5	S17	195	5/5	−42.8	223.0	243	4.9		
5	S18	192	4/4	−48.5	219.0	284	5.5		
5	S19	188	4/4	−52.1	222.8	873	3.1		
5	S20	184	5/5	−49.6	223.3	1117	2.3		
5	S21	181	4/4	−51.0	225.7	1023	2.9		
5	S22	176	6/6	−51.6	219.8	436	3.2		
5	S23	169	6/6	−53.2	222.8	588	2.8		
5	S24	165	5/5	−50.2	220.0	545	3.3		
5	S25	158	6/6	−49.9	215.0	2771	1.3		
5	S26	148	6/6	−47.8	214.4	340	3.6		
DG5 mean			**51/51**	**−49.7**	**220.3**	**274**	**1.2**	**−49.4**	**294.2**
6	S27	139	4/4	−51.1	195.7	192	6.7		
DG6 mean			**4/4**	**−51.1**	**195.7**	**192**	**6.7**	**−58.5**	**326.9**
7	S28	135	5/5	−66.1	188.0	1268	2.2		
DG7 mean			**5/5**	**−66.1**	**188.0**	**1268**	**2.2**	**−76.1**	**330.4**
8	S29	132	6/6	−63.6	185.4	1222	1.9		
DG8 mean			**6/6**	**−63.6**	**185.4**	**1222**	**1.9**	**−73.2**	**339.8**
9	S30	125	6/6	−60.6	192.7	1760	1.6		
9	S31	117	5/5	−59.3	190.3	148	6.3		
DG9 mean			**11/11**	**−60.1**	**191.7**	**320**	**2.6**	**−68.2**	**328.8**
10	S32	114	5/5	−71.4	200.6	706	2.9		
10	S33	112	6/6	−72.4	204.7	294	3.9		
10	S34	108	4/4	−71.5	193.0	663	3.6		
DG10 mean			**15/15**	**−71.9**	**200.1**	**432**	**1.8**	**−78.7**	**279.2**
11	S35	93	6/6	−45.3	171.9	267	4.1		
11	S36	90	5/5	−43.5	174.6	343	4.1		
11	S37	86	5/5	−39.7	175.1	146	6.4		
DG11 mean			**16/16**	**−43.0**	**173.7**	**205**	**2.6**	**−53.1**	**2.6**

Table 1a (*Continued*).

DG #	Site	Alt. (m)	*n*/*N*	Inc. (°)	Dec. (°)	*k*	α_{95} (°)	VGP Lat. (°N)	VGP Long. (°E)
Breidadalsá									
12	B1	375	7/7	−54.8	177.5	362	3.2		
12	B2	370	6/6	−51.3	176.8	654	2.6		
12	B3	365	5/5	−50.6	171.4	316	4.2		
12	B4	360	8/8	−50.7	175.9	692	2.1		
DG12 mean			**26/26**	**−52.5**	**177.9**	**88**	**3.0**	**−60.9**	**356.6**
13	B5	356	5/5	−68.5	193.8	244	4.9		
DG13 mean			**6/6**	**−68.5**	**193.8**	**244**	**4.9**	**−77.3**	**311.0**
14	B6	352	6/6	−65.3	217.3	439	3.2		
14	B7	350	7/7	−67.9	208.1	283	3.6		
14	B8	345	5/5	−69.7	208.3	391	3.9		
DG14 mean			**19/19**	**−67.7**	**212.9**	**246**	**2.1**	**−68.8**	**280.6**
15	B9	340	5/5	−60.6	183.6	495	3.4		
15	B10	336	4/4	−64.7	185.2	659	3.6		
DG15 mean			**9/9**	**−62.4**	**184.3**	**439**	**2.5**	**−71.4**	**343.2**
16	B11	330	5/5	−57.9	201.5	154	6.2		
16	B12	327	4/4	−52.4	208.9	443	4.4		
DG16 mean			**9/9**	**−55.6**	**205.1**	**164**	**4.0**	**−59.7**	**310.3**
17	B13	320	6/6	−41.8	207.5	340	3.6		
DG17 mean			**6/6**	**−41.8**	**207.5**	**340**	**3.6**	**−47.7**	**314.3**
18	B14	314	5/5	−54.2	201.9	271	4.7		
DG18 mean			**5/5**	**−54.2**	**201.9**	**271**	**4.7**	**−59.3**	**316.1**
19	B15	309	5/5	−46.2	213.2	307	3.8		
DG19 mean			**5/5**	**−46.2**	**213.2**	**307**	**3.8**	**−49.3**	**305.8**
20	B16	304	5/5	−52.5	207.0	312	4.3		
20	B17	299	6/6	−52.4	209.2	384	3.4		
DG20 mean			**11/11**	**−52.8**	**208.4**	**389**	**2.3**	**−56.1**	**307.7**
21	B18	294	8/8	−60.6	191.8	293	3.2		
DG21 mean			**8/8**	**−60.6**	**191.8**	**293**	**3.2**	**−68.3**	**328.6**

S is for Sandvík, B is for Breidadalsá. Alt. (m) is the altitude of the site in meters; *N* is number of treated samples; *n* is number of samples used for calculation; Inc., Dec. are the inclination and declination of the tectonically corrected site-mean ChRMs; *k*, α_{95} are the precision parameter and the confidence cone of the Fisher statistics [30]; VGP (Lat., Long.) are the latitude and longitude of the VGP. The geographic coordinates are: 61.64°N, 353.06°E for Sandvík; 62.13°N, 353.02°E for Breidadalsá.

deposits, which are followed by thin, dominantly olivine–phyric, or aphyric flow units, which are in turn overlain by dominantly plagioclase–phyric flow units [19,23]. The exposed basalts have undergone low-grade burial metamorphism in the chabazite–thomsonite to stilbite–heulandite zeolite zones suggesting paleotemperatures from less than 70°C and up to 110°C (O. Jørgensen, personal communication).

For the present study we sampled four separate profiles (86 lava flows, 548 drill cores), the stratigraphic and geographic positions of which are shown on Fig. 1b. The lower formation was sampled along two partly overlapping profiles, Vágur and Fámjin (profiles 1 and 2 on Fig. 1b). The much thinner flows of the middle formation at both Sandvík and Breidadalsá profiles (profiles 3 and 4 on Fig. 1b) made sampling of each flow unfeasible and consequently only every third to fifth flow was drilled. At Vágur the lava pile dips about 6°, at Fámjin and Sandvík 4°, while at Breidadalsá the flows are close to horizontal with a dip of 2°. All paleomagnetic directions and virtual geomagnetic poles (VGPs) reported in the following are in stratigraphic coordinates, i.e. corrected for tectonic dips. The cores were drilled directly in the field with a portable gasoline-powered drill and were oriented using both magnetic and sun compass. In the laboratory the cores were subsampled into standard cylindri-

cal specimens (in general, two to four specimens per core).

3. Results

In order to isolate the characteristic remanent magnetization (ChRM) both alternating field (AF) and thermal step-wise demagnetizations were performed. Thermal demagnetization was performed at l'Université Montpellier II in a magnetic shielded room with a CTF cryogenic magnetometer or a JR5a spinner magnetometer. The AF demagnetizations were performed at the University of Lund using a 2G cryogenic magnetometer with automatic three-axis AF demagnetiza-

Table 1b
Paleodirectional data for two profiles of the lower formation

Cooling unit #	Site	Alt. (m)	*n*/*N*	Inc. (°)	Dec. (°)	*k*	α_{95} (°)	VGP Lat. (°N)	VGP Long. (°E)
1	V1	0	6/6	34.5	325.6	72	7.9	−41.2	38.3
2	V2	10	6/6	62.0	35.7	334	3.7	−62.2	287.5
3	V3	25	5/5	59.7	359.0	256	4.8	−69.1	355.3
4	V4	55	6/6	74.2	357.2	672	2.6	−88.3	48.9
5	V5	65	8/8	72.0	335.1	243	3.6	−76.6	74.8
6	V6	100	0/6	*lightning*					
7	V7	140	10/12	71.2	28.5	357	2.6	−74.3	271.5
8	F1	10	4/4	−69.4	177.6	147	7.6	−81.4	2.9
	V8	*305*	*6/6*	*−75.4*	*190.3*	*417*	*3.3*		
	F2	*25*	*10/10*	*−74.2*	*205.1*	*565*	*2.0*		
9	**V8+F2**		**16/16**	**−74.8**	**199.9**	**404**	**1.8**	**−80.5**	**254.7**
	V9	*330*	*6/6*	*−72.7*	*199.3*	*477*	*3.1*		
	F3	*50*	*7/7*	*−70.4*	*194.8*	*174*	*4.6*		
10	**V9+F3**		**13/13**	**−71.2**	**196.5**	**238**	**2.7**	**−79.7**	**289.8**
	V10	*365*	*6/7*	*−64.4*	*177.8*	*304*	*3.8*		
	F4	*75*	*0/6*	*lightning*					
11	**V10+F4**		**6/13**	**−64.4**	**177.8**	**304**	**3.8**	**−74.6**	**358.9**
	V11	*410*	*5/5*	*−54.1*	*185.8*	*183*	*5.7*		
	F5	*125*	*7/8*	*−62.0*	*191.9*	*41*	*9.6*		
12	**V11+F5**		**12/13**	**−58.7**	**189.0**	**55**	**5.9**	**−67.2**	**334.9**
	V12	*450*	*2/6*	*−38.5*	*168.0*				
	F6	*165*	*9/9*	*−42.5*	*169.9*	*75*	*6.0*		
13	**V12+F6**		**11/15**	**−41.7**	**169.5**	**76**	5.3	−51.8	8.7
	V13	*465*	*3/6*	*−44.3*	*183.7*	*196*	*8.8*		
	F7	*180*	*6/6*	*−55.6*	*174.0*	*357*	*3.5*		
14	**V13+F7**		**9/12**	**−51.9**	**177.7**	**108**	**5.0**	**−60.9**	**357.1**
	V14	*480*	*0/8*	*lightning*					
	F8	*192*	*6/6*	*−62.7*	*185.2*	*286*	*4.0*		
15	**V14+F8**		**6/14**	**−62.7**	**185.2**	**286**	**4.0**	**−72.3**	**340.8**
16	F9	215	4/7	55.8	355.7	10 436	0.9	−64.7	1.2
17	F10	240	8/8	52.3	348.4	191	4.0	−60.4	13.1
18	F11	271	8/8	89.0	94.1	168	4.3	−61.3	177.3
19	F12	300	7/7	72.9	317.0	270	3.7	−68.7	94.0
20	F13	310	9/9	68.9	358.9	353	2.7	−80.8	357.3
21	F14	315	8/8	63.4	8.3	283	3.3	−72.7	333.0
22	F15	365	8/8	38.7	30.5	194	4.0	−45.1	311.3
23	F16	375	7/7	−39.3	173.8	247	3.8	−50.5	2.2

V is for Vágur, F is for Fámjin. Results of lava flows sampled at both profiles (in italics) were combined to obtain a mean for the given cooling unit (in bold). The geographic coordinates are: 61.47°N, 353.16°E for Vágur; 61.53°N, 353.12°E for Fámjin. Notations are the same as in Table 1A.

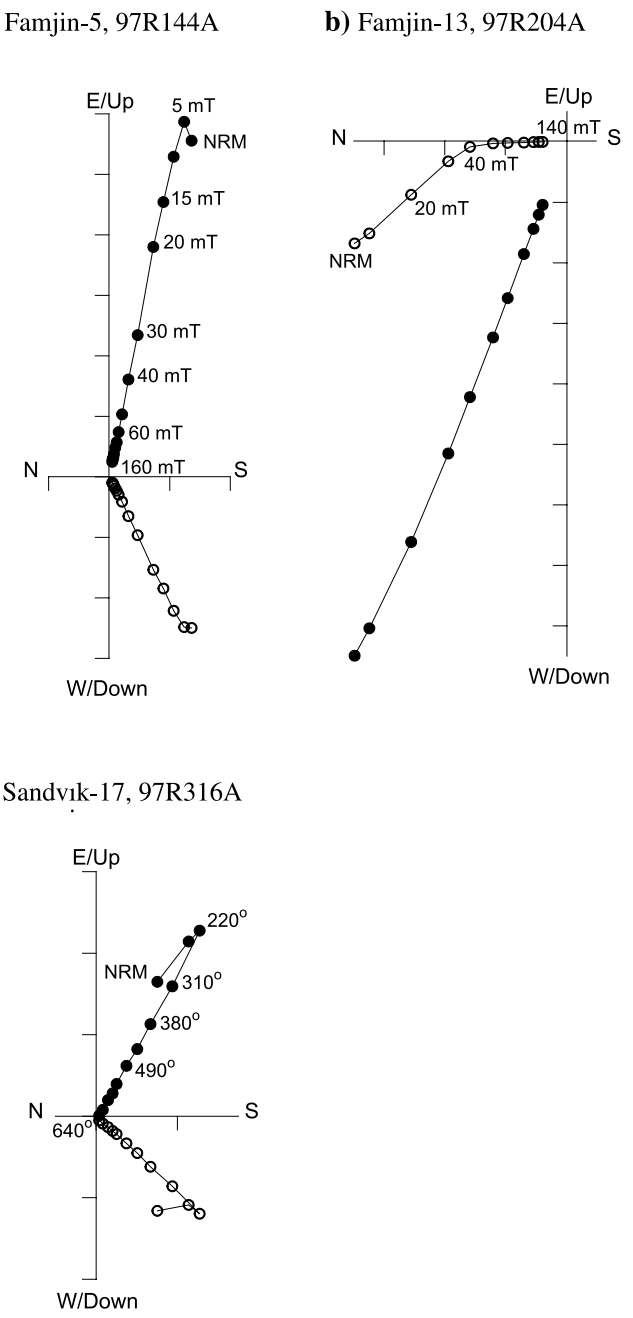


Fig. 2. Examples of orthogonal projections and NRM decay of AF (a and b) and thermal demagnetizations (c). Solid (open) symbols on the orthogonal projection correspond to vertical (horizontal) planes.

tion. The direction of the ChRM was calculated using standard principal component analysis [24] accepting data with maximum angle of deviation less than 2°. Typical demagnetization results are shown in Fig. 2. It is important to note that while the majority (~70%) of the samples showed one-component NRM demagnetizations (Fig. 2a), some 30% contained significant secondary compo-

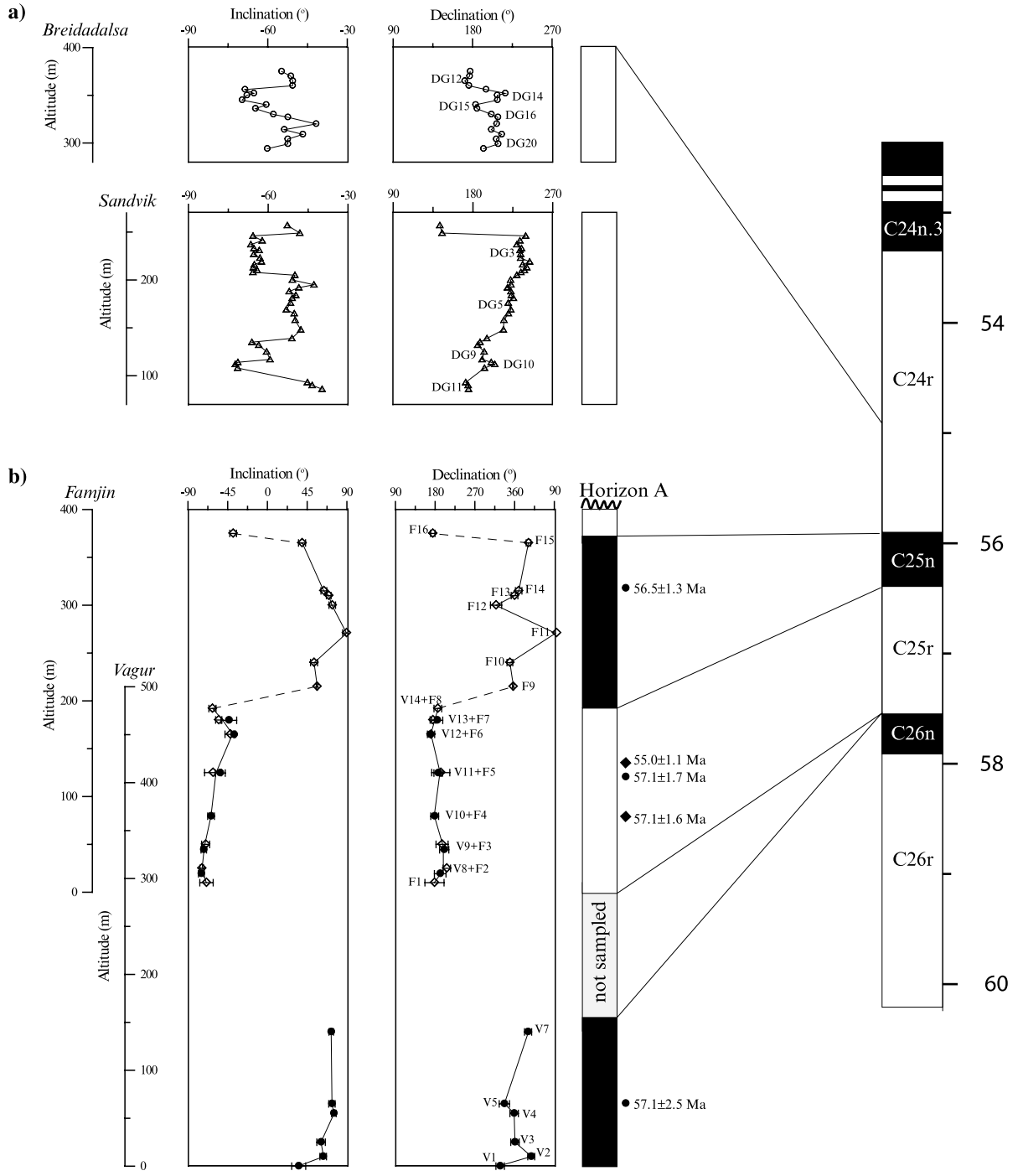


Fig. 3. Direction of ChRM in composite profile through the middle formation (a) and the lower formation (b) of the volcanic succession. Closed circles are for Vágur, diamonds are for Fámjin. Shown to the right is the magnetic polarity. Circles indicate whole-rock K/Ar ages; diamonds are whole-rock $^{40}Ar/^{39}Ar$ ages [27]. The correlation to the GPTS [38] is based on high-precision $^{40}Ar/^{39}Ar$ dating on mineral separates with ages of 59–56 Ma for the lower formation and 56–55.5 Ma for middle and upper formations [23,26].

nents, most likely of viscous origin. For these samples the ChRM often could not be defined before AF demagnetization to ~40 mT (Fig. 2b) or thermal demagnetization to ~300°C (Fig. 2c). Finally, three sites of the lower formation (Vágur-6 and -14, Fámjin-4) had highly scattered NRM directions and demagnetization results suggesting a lightning-induced isothermal overprint. It was not possible to isolate the primary magnetization for these three sites (see Table 1b) and they are excluded from further analysis.

3.1. Magnetostratigraphy

The site-mean paleomagnetic directions are summarized in Table 1a and Fig. 3afor the middle formation and Table 1b and Fig. 3b for the lower formation. For the middle formation an obvious characteristic is the presence of consecutive flows having statistically indistinguishable site-mean directions (Fig. 3a). We interpret this to indicate that these flows were erupted in a short time interval and hence recording essentially a spot reading of the Earth's magnetic field. To determine whether or not the paleomagnetic directions of consecutive lava flows are statistically distinct at the 95% confidence level, the test suggested by Watson [25] was used. Following the Watson test, successive directional groups (DGs) were defined as series of lavas with indistinct mean directions (Table 1a). From the altogether 55 flows sampled in the middle formation, we have obtained 21 DGs, several of which are remarkably thick, up to 50 m (~50 lava flows).

The two lower formation profiles are partly overlapping (Fig. 1b) and several flows were therefore sampled along both profiles. The flow-to-flow correlation suggested in Table 1b and Fig. 3b is based on direct field observations as well as chemical composition. The paleomagnetic directional data agree excellently with the proposed correlation and provide independent support that the very same flows have been sampled at the two localities. In the following analysis we combine ChRM directions so that each lower formation lava flow is represented only once (Table 1b).

High-precision $^{40}Ar/^{39}Ar$ dating on mineral separates from Faroes basalts yielded ages of ~59–56 Ma for the lower formation, and ~56–55.5 Ma for middle and upper formations [23,26]. On the other hand, whole-rock K/Ar and $^{40}Ar/^{39}Ar$ experiments on Faroes basalts were problematic due to K-containing clay minerals occurring interstitially in the groundmass, but also forming vesicle fillings or replacing olivine or, more rarely, plagioclase. The whole-rock ages shown on Fig. 3 are the ones deemed most reliable (for further details see [27]). Following the analysis of Waagstein [19] as well as the above mentioned age information the Faroes flood basalts can be robustly tied to the GPTS in the following manner (see Fig. 3): the lowermost exposed lavas were erupted during Chron 26n followed by at least eight lava flows of reverse polarity C25r, which are overlain by seven flows of normal polarity C25n. The last lava flow of the lower formation sampled right underneath the coal-bearing horizon A (Fig. 1) is of reverse polarity and may be correlated to magnetochron C24r. The middle and upper formations are reversely magnetized (see also [28]) and correlate to C24r.

3.2. Paleosecular variation

In order to detect and reject possible anomalous VGPs due to intermediate, excursional or erroneous site-mean directions, we used the method of Vandamme [29] to define a cut off angle of 43.5°. We find that all 43 lava flows/DGs (Tables 1a and 1b) lie within this cut off angle and hence all data are included in the following analysis. The paleosecular variation is usually expressed by the angular standard deviation (ASD) of the VGP distribution with the implicit assumption that the distribution is Fisherian [30]. In order to test whether our 43 VGPs are Fisher-distributed we produce quantile–quantile plots for the exponential colatitude and the uniform longitude (Fig. 4). Both plots are seen to be more or less linear, and with M_u and M_e values lower than 1.207 and 1.094 respectively; a Fisher distribution hypothesis cannot be rejected at the 95% confidence level [31].

Following McFadden et al. [32], we corrected

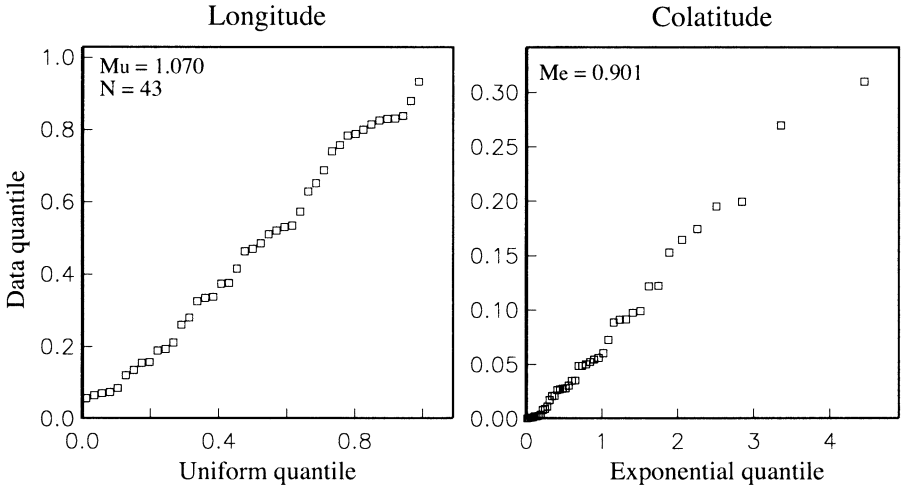


Fig. 4. Quantile–quantile plots assessing goodness-of-fit of the Fisher distribution to the Faroes VGPs. Values M_u are lower than 1.207 and M_e is lower than 1.094, hence the hypothesis of Fisher distribution cannot be rejected.

the ASD for an average within-site dispersion, $S_w = 4.7°$, with an average number of samples per lava flow/DG, $n = 12$. The ASD for the whole data set is 21.7° with a 95% confidence limit of +3.9°/−2.8° [33]. This value of ASD is a little higher, but within error of the value (~17°) predicted by the PSV model of McFadden et al. [32] for the given age and paleolatitude.

3.3. Paleomagnetic pole

The 43 individual paleomagnetic directions and their corresponding VGPs are shown in Fig. 5. As both normal and reverse polarities are present, a reversal test [34] was performed. The angle between mean normal and reverse polarity directions is 8.0°. The critical angle, for which the two directional data sets would be different at the 95% significance level, is 9.7°, thus, the collection passes the reversal test at level B [34]. The mean direction based on all 43 independent readings of the paleomagnetic field is inclination = 60.9°, declination = 7.7° ($\alpha_{95} = 4.5°$, $K = 24.5$). The mean paleomagnetic pole has coordinates 71.4°N, 154.7°E ($A_{95} = 6.0°$, $K = 14$, $N = 43$) (Fig. 5b). The magnetostratigraphy with four polarity zones as well as the paleosecular variation analysis clearly demonstrate that the paleomagnetic pole represents a time-averaged field.

4. Discussion

4.1. Magnetostratigraphy and timing of NAIP volcanism

The Faroes flood basalt succession has recently been correlated with the basalts in East Greenland [23] suggesting that volcanism was synchronous throughout the central part of the NAIP. Compilations of radiochronological ages from both onshore and offshore samplings propose that NAIP volcanism occurred in two major phases: phase 1 with ages falling in the range ~62–58 Ma and phase 2 with ages ~56–54 Ma [35,36]. In Faroe Islands, the volcanic hiatus between the two phases is marked by the sedimentary horizon A ([23]; Figs. 1b and 3). The well constrained Faroes magnetostratigraphy demonstrates that both the end of phase 1 and the onset of phase 2 volcanism occurred in C24r (Fig. 3), which is in conflict with the suggested onset of phase 2 in C25r [36] or C25n [35]. We speculate that the ~1–2 Ma discrepancy between our magnetostratigraphic data and the radiochronological ages relate to uncertainties in the whole-rock $^{40}Ar/^{39}Ar$ ages compiled by Saunders et al. [35] and Tegner et al. [36]. For example, whole-rock $^{40}Ar/^{39}Ar$ studies on several slightly altered Faroes basalts produced ages ~1–2 Ma too high although the

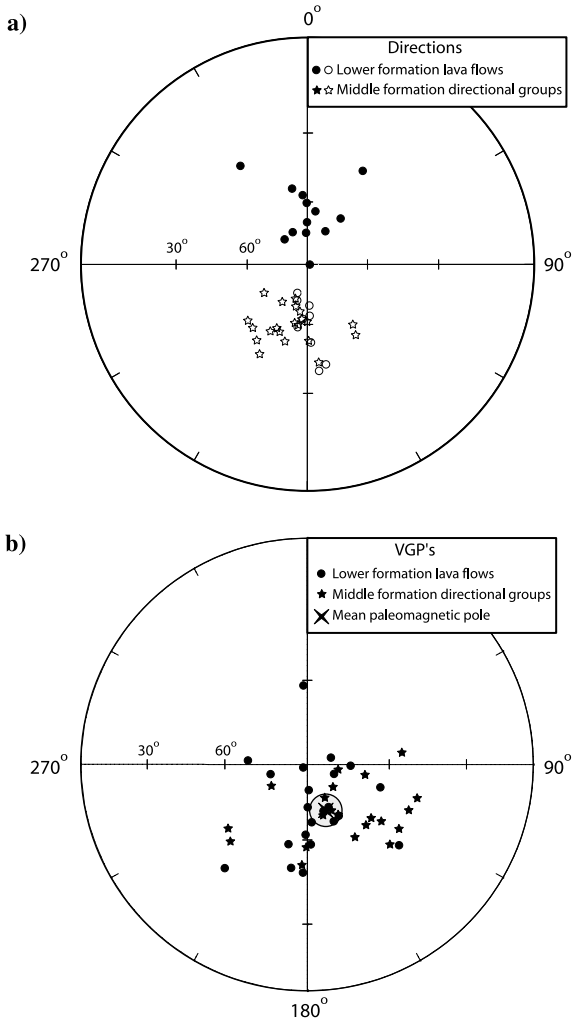


Fig. 5. (a) Stereo-diagram of mean directions for the lower formation lava flows and middle formation DGs. Open (solid) symbols correspond to negative (positive) inclination. (b) Equal-area diagram of corresponding VGPs. The mean paleomagnetic pole is shown by a cross, and the 95% confidence limit is also given.

age plateaus fulfil standard criteria of acceptance [27]. A similar observation has recently been made on Deccan trap volcanics [37]. Another source of discrepancy could be errors in the absolute ages of the GPTS [38] using a 55 Myr calibration point that may be problematic, resulting in a too young C25n–C24r polarity boundary [39].

Our new data place the onset of phase 2 volcanism in the same magnetochron as the North Atlantic break-up where the oldest ocean-floor magnetic lineation is C24r [2]. Larsen and Saunders [3] suggest that the oldest sea-floor anomaly offshore East Greenland is cryptochron C24r.11, i.e. the early part of C24r. Continental break-up and decompression melting would also explain the extreme accumulation rate at the very onset of phase 2 volcanism, evidenced by the ca. 50 m thick (~50 flow units) DG3 and DG5 of the middle formation Sandvík profile (Fig. 3a). Assuming that Paleogene rates of secular variation were similar to the well estimated Holocene rates, a DG would correspond to a time period of less than a few hundred years. At the Breidadalsá profile (Fig. 3a), 1000 m higher in the middle formation, the DGs are shorter, which may indicate a generally lower accumulation rate than at the onset of phase 2. In contrast, the magnetostratigraphy of the Faroes lower formation (Fig. 3b) points to a significant slowdown in the effusive activity towards the end of phase 1 possibly reflecting the effect of a thickening volcanic crust. An exact estimate can be obtained based on the seven lavas in the 487 kyr long Chron C25n [38] yielding an eruption rate of about one flow per 70 kyr.

Rapid global warming coincides with at least six out of the 11 intervals of large igneous province formation discussed by Wignall [40]. However, exact correlation of flood volcanism and paleoclimate change is made difficult due to poor age control, unknown eruption rates and therefore highly uncertain estimates of the annual volcanic CO_2 output. Cyclostratigraphic data indicate that the $\delta^{13}C$ anomaly associated with the late Paleocene thermal maximum began 924–966 kyr into C24r [4] and had a duration of 210–220 kyr [5]. The $\delta^{13}C$ excursion is too large to be accounted for by release of volcanic CO_2 alone, instead methane hydrate is widely regarded to have supplied the light C. The $\delta^{13}C$ excursion was, however, preceded by a ~30 kyr period of gradual climate warming [5]. Based on the temporal coincidence and the high eruption rate of phase 2 volcanism, it is possible that NAIP volcanic CO_2 could have initiated greenhouse warming that triggered the dissociation of isotopically light methane hydrates.

4.2. Paleomagnetic pole and reliability of previously published NAIP results

The Eurasian part of the NAIP has been extensively studied with 23 British and four Faroes paleomagnetic poles listed in the on-line IAGA global paleomagnetic database (http://dragon.ngu.no/Palmag/paleomag.htm) updated in 2001. These poles are, however, of varying quality and none of them is based on principal component analysis, i.e. all have a demagnetization code equal to or less than 3 in the McElhinny and Lock [41] terminology. Following Torsvik et al. [14] and Van der Voo and Torsvik [13] and rejecting poles with a quality factor [42] lower than 3, only 13 British and two Faroes poles remain (Table 2; Fig. 6a). Although the 15 selected poles are the apparently most reliable, a problem is seen already from the fact that several poles have 95% confidence limits that do not overlap and hence they are statistically different at the 95% confidence level (Fig. 6a). Seven out of 15 poles are statistically different from the new Faroes pole presented in this study (Fig. 6a). We believe that this inconsistency reflects the outdated analytical and laboratory techniques used in the older NAIP studies. Note, for example, that the two selected Faroes paleomagnetic poles [28,43] are based on blanket AF demagnetization that would lead to an erroneous result for samples with a strong viscous component like the one depicted in Fig. 2b (30% of the samples in our study). The interpretation of Tarling [28], that the non-Fisherian VGP distribution observed in his Faroes study depicts a true geomagnetic phenomenon, is at odds with our results (Fig. 4) and therefore more likely represents a non-random bias caused by unremoved viscous overprints. In general, it is difficult to discuss the reliability of the older NAIP poles as necessary details, such as, for example, site-mean

Table 2
Selected paleomagnetic poles from Eurasian part of NAIP

Rock formation	Age (Ma)	Plat (°)	Plong (°)	A_{95} (°)	K	B	N	ASD (°)	Q	Demag code	GPDB #
Lundy island dykes	49.5	83	155	1.5	–	48	155	6.0	5	2	755
Dykes and contacts	50	73	197	4.0	17	21	84	10.6	6	2	340
Dyke swarm, Skye	51	82	158	2.0	–	409	1636	23.4	4	2	1174
Fishnish dykes	52.5	74	139	10.0	15	15	57	22.4	5	2	1040
Vaternish dykes	55	76	160	3.5	21	81	324	18.2	5	2	85
Antrim basalts	56	70	163	5.0	33	25	225	14.5	4	2	654
Faroe Islands basalts	57	80	189	6.0	14	33	33	19.9	3	3	419
Faroe Islands volcanics	57	77	161	1.7	–	253	1809	15.6	3	2	650
Skye Lavas	57.5	72	165	2.5	36	90	344	13.7	3	2	86
Muck and Eigg volcanics	58	74	171	2.7	22	133	524	18.0	5	3	1204
Rhum and Canna volcanics	59	81	179	2.4	34	107	445	14.4	6	2	1169
Ardnamurchan complex	59.5	77	175	2.7	-	62	484	12.3	4	3	1377
Mull dykes	60.5	78	187	2.7	17	154	530	19.4	7	3	83
Arran dykes	61	82	180	1.2	37	413	413	14.1	5	2	1041
Mull lavas	62	72	168	2.8	32	78	492	14.3	5	2	1055
Mean pole		77	164	2.6							

Data are selected from the global paleomagnetic database (GPDB; see text). Age (Ma) is the mean age listed in the GPDB; Plat (°) and Plong (°) are the latitude and longitude of the paleomagnetic pole; A_{95} (°) and K are the confidence cone and the precision parameter of the Fisher statistics [30]; B is the number of sites; N is the number of samples; ASD (°) is the estimated angular standard deviation (calculations based on A_{95} and B after formulas (6.18) and (6.22) in Butler [49]); Q is the quality factor [42]; Demag code is the demagnetization code in the McElhinny and Lock [41] terminology. Mean of the 15 selected poles is given.

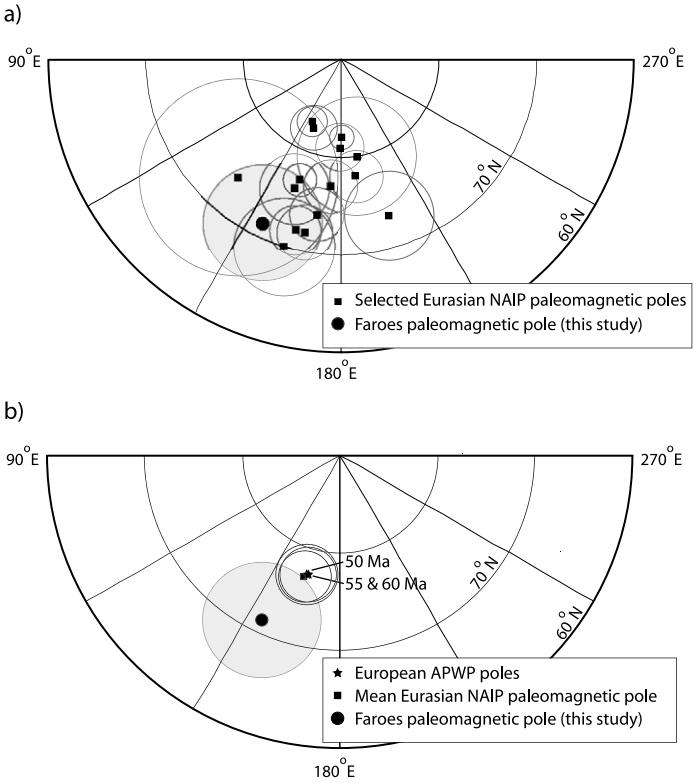


Fig. 6. Equal-area projection of Faroes mean paleomagnetic pole obtained in this study together with (a) 15 selected Eurasian NAIP paleomagnetic poles (see Table 2); (b) the mean Eurasian NAIP pole (Table 2) and the 50, 55 and 60 Ma APWP poles [14].

directions, which are most often not listed. An estimate of the ASD can, however, be deduced based on the number of cooling units and the A_{95} (Table 2). The ASD values are remarkably low for some of the poles, with eight poles having ASD values lower than 15° and hence statistically lower than the ASD of our new Faroes data set. It is therefore highly questionable if secular variation has been sufficiently averaged in these studies.

In Fig. 6b the new Faroes paleomagnetic pole is shown together with the mean Eurasian NAIP pole (Table 2) as well as the 50, 55 and 60 Ma poles of the most recently published apparent polar wander path (APWP) for Europe based on running means with a 20 Myr moving window [14]. The 50–60 Ma APWP is based almost solely on the Eurasian NAIP data, where, for example, 15 out of 16 poles in the 55 Ma APWP pole are from the Eurasian NAIP. It is therefore not surprising that APWP poles and the mean Eurasian NAIP pole are indistinguishable (Fig. 6b). Finally it is worth mentioning that ages of the Eurasian NAIP listed in the global paleomagnetic database are taken from the original publications (Table 2), some of which have since been revised. It is now well established that the bulk of volcanism on the British Isles was confined to the period 61–59 Ma, whereas late stage dyke emplacement continued until ~58 Ma [44], implying that many of the original ages are too young (see Table 2).

5. Conclusions

Based on the Faroes magnetostratigraphy we suggest that the temporal NAIP frame work offered by whole-rock $^{40}Ar/^{39}Ar$ ages are slightly

offset towards too old ages. The magnetostratigraphic data suggest a close temporal relation between the onset of NAIP phase 2 volcanism, the North Atlantic continental break-up, and the late Paleocene thermal maximum. The extreme increase in accumulation rate at the beginning of phase 2 most likely reflects extensional processes during final North Atlantic continental break-up. Prior to the break-up we obtained an exact estimate of the eruption rate of 1/70 kyr for C25n.

Of the altogether 8639 paleomagnetic poles listed in the on-line IAGA global paleomagnetic database more than 75% stem from studies performed before 1980 and the advent of principal component analysis, i.e. having a demagnetization code equal to or less than 3 in the McElhinny and Lock [41] terminology. The importance of high quality paleomagnetic poles are underlined by recent debate regarding suggested true polar wander events at 84 Ma [45,46], and ~520 Ma [47,48], where widely different conclusions are being made following different data selection criteria. Field tests and paleomagnetic quality factor, Q [42], are important for older rocks for which the primary nature of NRM is uncertain. However, as shown by the present study, detailed step-wise thermal and AF demagnetization and principal component analysis are of vital importance if fine-scale features of the Earth's paleomagnetic field, APWP or TPW are to be tested. We suggest that our Faroes paleomagnetic pole (71.4°N, 154.7°E; $A_{95} = 6.0°$, $K = 14$, $N = 43$) supersedes the NAIP poles from previous less reliable studies.

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